

ABHINAV CHOWDARY KORITALA

abhinavkoritalattu@gmail.com | +1 (806)-730-1021 | www.linkedin.com/in/abhinavkoritala

EDUCATION

MS in COMPUTER SCIENCE | TEXASTECH UNIVERSITY

Aug 2023 - May 2025

BACHELOR OF TECHNOLOGY, COMPUTER SCIENCE | VFSTR UNIVERSITY | 8.3

Aug 2019 - May 2023

Coursework: Data Structures, Database Management Systems, Computer Organization and Architecture, Design and Analysis of Algorithms, Web Technologies, Python Programming, Compiler Design, Operating Systems, Computer Networks, Artificial Intelligence, Data Science using Python, Cloud Computing, Machine Learning, Big Data Analytics.

TECHNICAL SKILLS

- **Programming & Scripting Languages** | C, Python, SQL, HTML, CSS
- **Data Analysis & Visualization** | Pandas, Numpy, Matplotlib, Seaborn, Sci-kit Learn, TensorFlow, Keras, OpenCV
- **Statistics & Machine Learning** | Statistical Analysis, Regression Analysis, Classification Algorithms, Clustering, Reinforcement Learning, Predictive Models, Dimensionality Reduction
- **Deep Learning** | Convolutional Neural Networks (CNN), Feature Extraction, Large Learning Model, PyTorch
- **Tools & Frameworks** | Django, Bootstrap, React, Jupyter-NB, Google Colab, MS, GIT Commands, VScode
- **Cloud Platforms & Operating Systems** | AWS, Windows, Unix
- **Soft Skills** | Teamwork, Problem-Solving, Quick Learning

PROJECTS

Movie Recommender System (Content-Based) | Streamlit, NLTK, Python, VScode

- Built an interactive movie recommendation system using Streamlit for a user-friendly interface.
- Cleaned and preprocessed movie data with Python, applying PorterStemmer for text normalization.
- Used CountVectorizer to transform movie descriptions into numerical feature vectors based on word frequency, enabling text-based comparisons.
- Calculated cosine similarity between movie descriptions to recommend similar movies based on content similarity.

Brain Tumor Prediction Using MRI Images | UNET, CNN

- Preprocessed over 10,000 MRI images containing diverse samples which enabled precise feature learning by the CNN architecture designed for automatic diagnosis assistance aimed at reducing misdiagnosis rates among patients with critical conditions.

Chat Bot using local LLM | Streamlit, FAISS, GPT4

- Developed a local LLM application utilizing GPT4All's Meta-Llama-3-8B-Instruct model and HuggingFace's all-MiniLM-L6-v2 Sentence Transformer to process user queries and generate relevant answers.
- Implemented a query-to-answer pipeline with document retrieval, context preparation, and citation display, all within a custom Streamlit UI.

Virtual Painter | Artificial Intelligence, Open CV

- Created a pioneering system that leveraged OpenCV for non-contact drawing, which was adopted by 15+ educational institutions, streamlining art and design curriculums while promoting innovative teaching methods.

CERTIFICATIONS & ACHIEVEMENTS

- **Programming in Python (Meta)** | [CERTIFICATE](#)
 - Completed the Meta Python Programming Certification Course, mastering Python programming, data structures, and software development best practices, with a focus on real-world applications.
- **Machine Learning A-Z: AI, Python Certification (Udemy)** | [CERTIFICATE](#)
 - Gaining hands-on experience in machine learning algorithms, data analysis, and model deployment using Python.
- **Introduction to Back-End Development (Meta)** | [CERTIFICATE](#)
 - Acquiring skills in server-side Programming, database management and API design.